

Supplement: matched perfect-correction Markov channel and continuous Lindblad campaign

Internal campaign summary

(Dated: Production supplement; completed August 22, 2026)

I. NOTATION AND LOCAL-MAP DERIVATION

This supplement uses the overcomplete Wannier (OW) conventions of Bhuiyan–Pan–Jian. For a normalized OW state $|\widetilde{W}_{\mathbf{r},\nu,n}\rangle$, set $P = |\widetilde{W}\rangle\langle\widetilde{W}|$ and $\hat{N} = \hat{\chi}^\dagger\hat{\chi}$. The number-dephasing adjoint action closes on the two-point correlation matrix:

$$\mathcal{D}^\dagger[\hat{N}](G) = PGP - \frac{1}{2}\{P, G\}. \quad (1)$$

Unit perfect filling and depletion contribute, respectively,

$$P - \frac{1}{2}\{P, G\}, \quad -\frac{1}{2}\{P, G\}. \quad (2)$$

Writing $\mathcal{I}$ for the identity superoperator, adding the number-dephasing term gives the exact filled and empty reset identities

$$\mathcal{R}_{P,-} - \mathcal{I} = \mathcal{D}^\dagger[\hat{\chi}^\dagger] + \mathcal{D}^\dagger[\hat{N}], \quad (3)$$

$$\mathcal{R}_{P,+} - \mathcal{I} = \mathcal{D}^\dagger[\hat{\chi}] + \mathcal{D}^\dagger[\hat{N}]. \quad (4)$$

Thus each exact reset has the form $\mathcal{R}_a = \mathcal{I} + \mathcal{L}_a$. The canonical within-cell order is $A+$ (loss), $A-$ (gain), $B+$ (loss), and $B-$ (gain). By contrast, $\mathcal{I} + \mathcal{D}^\dagger[\hat{\chi}]$ and $\mathcal{I} + \mathcal{D}^\dagger[\hat{\chi}^\dagger]$ at unit coefficient are formal Euler increments, not exact completely positive reset maps. Expanding a product of weakened reset maps $\mathcal{R}_P(p) = \mathcal{I} + p[\mathcal{R}_P - \mathcal{I}]$ gives

$$\prod_a \mathcal{R}_{P_a}(p) = \mathcal{I} + p \sum_a [\mathcal{R}_{P_a} - \mathcal{I}] + O(p^2), \quad (5)$$

which fixes the continuous generator and shows why its first derivative is independent of site order even though the finite product is not. For the physical unit-step sweep, $\mathcal{E}_s = \mathcal{R}_M \cdots \mathcal{R}_1 = \mathcal{I} + \mathcal{L}_{\text{cycle},s}$ is also an exact identity, but $\mathcal{L}_{\text{cycle},s} = \mathcal{E}_s - \mathcal{I}$ contains schedule-ordered cross terms. The matched infinitesimal generator is instead $\sum_a \mathcal{L}_a$ (or the corresponding dephasing-deleted sum).

II. TRANSLATION TWIRL AND GLOBAL MOMENTUM WEIGHTS

The spatial sliding estimator is the finite-group conditional expectation

$$\mathcal{T}_y[G] = \frac{1}{N_y} \sum_{q=0}^{N_y-1} T_y^q G T_y^{-q}. \quad (6)$$

It is Hermiticity preserving, trace preserving, positive, idempotent, and exactly translation invariant. These properties are tested directly. They do not imply that the untwirled ordered channel commutes with T_y .

For global eigenmodes, Parseval normalization gives

$$\sum_{k_y} Z_j(k_y) = \sum_{x,y,\mu} |\phi_j(x, y, \mu)|^2 = 1. \quad (7)$$

The archive retains both the global occupations and $Z_j(k_y)$ so that a momentum-weighted spectral function can be reconstructed without assuming symmetry.

III. EXACT CASE MATRIX

The configuration expands to 105 unique case specifications. Endpoint scans include DW-support truncation on and off; alpha scans include truncation on only. Counts are

Dynamics	Dephasing	Cases
Ordered Markov channel	on	27
Continuous Lindblad	off	39
Continuous Lindblad	on	39

One of the 27 channel case specifications is a grouped $S = 10$ schedule-seed control; the main-grid copy of the same physical parameters retains its single matched seed. Response is computed for exactly 22 endpoint cases: finite $n_{\text{shell}} = 1$ at $N_y = 32, 48, 64$ with truncation on/off for each of the three arms, and full-frame continuous cases at $N_y = 64$ with truncation on/off. Alpha scans, $n_{\text{shell}} = 2$, and the grouped schedule control intentionally omit response arrays.

IV. ARRAY-LEVEL STORAGE CONTRACT

Every cycle-resolved observable has leading logical axes (s, c) , with exact sample IDs and $c = 0, 1, \dots, 2N_y$. Only inexpensive traces, norms, and translation diagnostics are evaluated at every cycle. Occupation spectra and entropy proxies are evaluated at the explicit checkpoint cycle list in the configuration. Final and late-average dense correlation matrices are checkpoint-resolved and retain their sample axis.

The response product retains `response_density_source_wall_time_y` with logical shape (s, y_0, a, t, y) before source averaging. The derived source-mean profile has shape (s, a, t, y) . Channel and continuous time coordinates are saved explicitly and are not silently treated as identical grids.

Each completed case directory contains exactly one compressed observable file, one JSON metadata file, and a success marker. A failed case has a failure record and failure marker. Temporary directories are renamed atomically; existing case and run roots are never overwritten. The run manifest pins configuration, solver, adapter, canonical class, reference-code, and git-state hashes.

V. VALIDATION MATRIX

Gate	Acceptance rule
Grid	Exactly 105 case specifications and arm counts 27, 39, 39.
Seeds	Every main-grid channel variant at fixed N_y has the same schedule-match key and seed; the grouped control has ten unique, replayable seeds.
Geometry	Both engines use $N_x = 20$, walls 5, 15, and identical shell/DW-mask recipes. Gauge-dependent raw-OW hashes are audited but are not treated as physical projector invariants; three warnings are retained in the analysis JSON.
Local algebra	Random physical small-system matrices reproduce all canonical local perfect-correction updates.
Continuous derivative	Central finite differences of weakened local maps converge to the implemented generator with both dephasing settings.
Physicality	Full-dephasing channel and both continuous arms obey $-\epsilon \leq \nu \leq 1 + \epsilon$ at all spectral checkpoints.
Translation	Continuous late states meet the configured exact-symmetry tolerance; channel residuals are recorded rather than required to vanish.
Twirl	Hermiticity, trace, positivity, idempotence, and translation invariance hold to complex128 tolerance.
Time grid	Every cheap observable contains every sample and cycle with no duplicates or gaps; spectral checkpoints match the declared sparse coordinate.
RK4	The representative $\delta t = 0.1, 0.05, 0.025$ comparison exhibits fourth-order global convergence before production.
Response	Analytic tangent and finite $\pm\epsilon$ central difference agree; source averaging occurs only after source-resolved retention.
Provenance	All source and result hashes match before and after analysis; the legacy archive remains byte-identical.

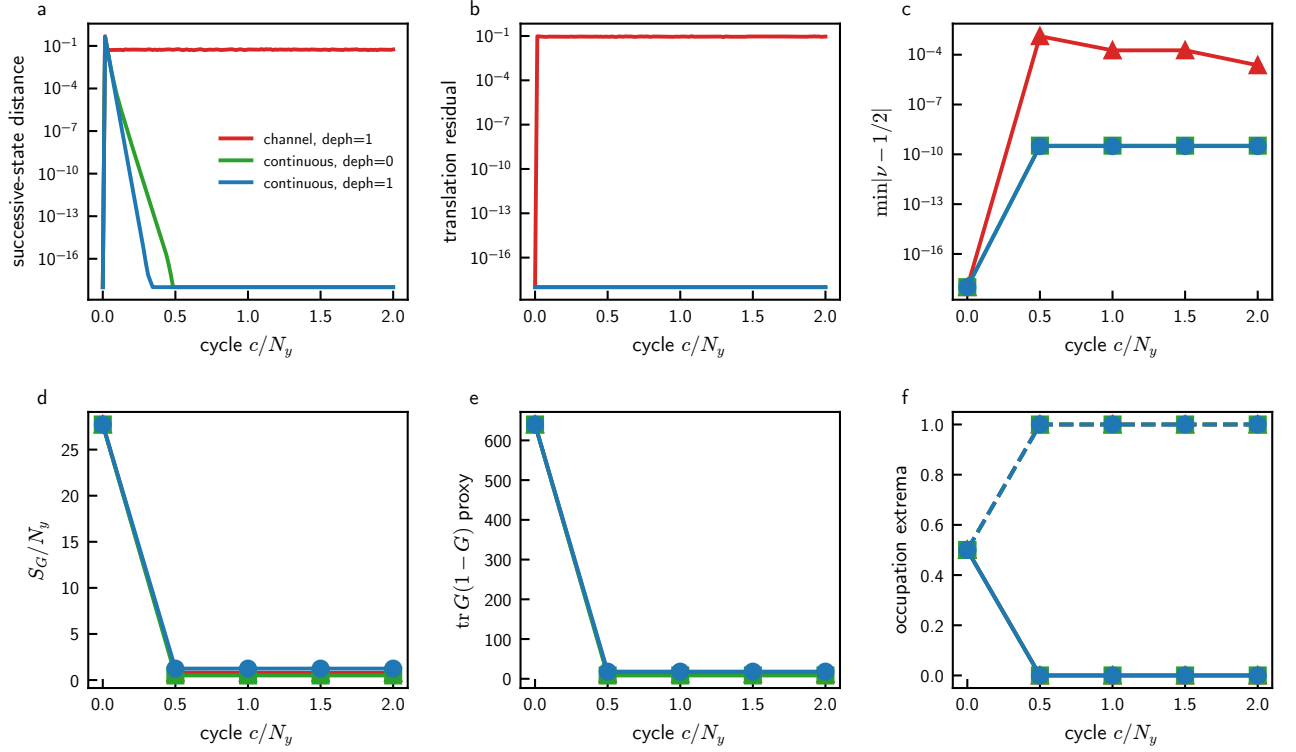


FIG. 1. Cycle diagnostics for the canonical $N_y = 64$, $n_{\text{shell}} = 1$, DW-truncated endpoint. The continuous arms reach machine-level successive-state and translation residuals by approximately $c/N_y = 0.5$. The random ordered channel retains a nonzero cycle-to-cycle distance and translation residual, so its reported stationary spectrum is explicitly based on the late temporal average and translation twirl. The twirl deletes off-diagonal momentum coherences and is not a schedule average; the main text separately plots the raw untwirled global eigenvalues in ascending order. The remaining panels show checkpoint midpoint gap, Gaussian entropy and charge-variance proxies, and occupation extrema.

Gate	Acceptance rule
Document	Main text is at most nine pages, with no overfull boxes, undefined references, or unembedded fonts.

VI. PRODUCTION CONVERGENCE AND SCHEDULE DIAGNOSTICS

VII. LEGACY RECONCILIATION

The prior stationary solver used dephasing-free gain and loss with equal coefficient $1/2$. The new dephasing-free perfect-correction continuous generator has equal unit coefficients. Hence

$$\mathcal{L}_{\text{old}} = \frac{1}{2}\mathcal{L}_{\text{PC}}^{(0)}, \quad G_{\text{ss,old}} = G_{\text{ss,PC}}^{(0)} \quad (8)$$

when the geometry and OW projectors match. State-only stationary estimators can be used as cross-checks. Relaxation rates, fixed-time response, and the old auxiliary Strang map cannot be relabeled as results of the requested dynamics.

VIII. CLAIM DISCIPLINE

A channel twirl is not an exact symmetry of a fixed ordered sweep. A natural-occupation crossing is not a zero Liouvillian decay rate. A signed density-lobe velocity is not a Hamiltonian group velocity or quantized Hall coefficient. Ten schedules quantify only the specified schedule ensemble at the canonical endpoint. These distinctions remain in captions, tables, and machine-readable analysis metadata.

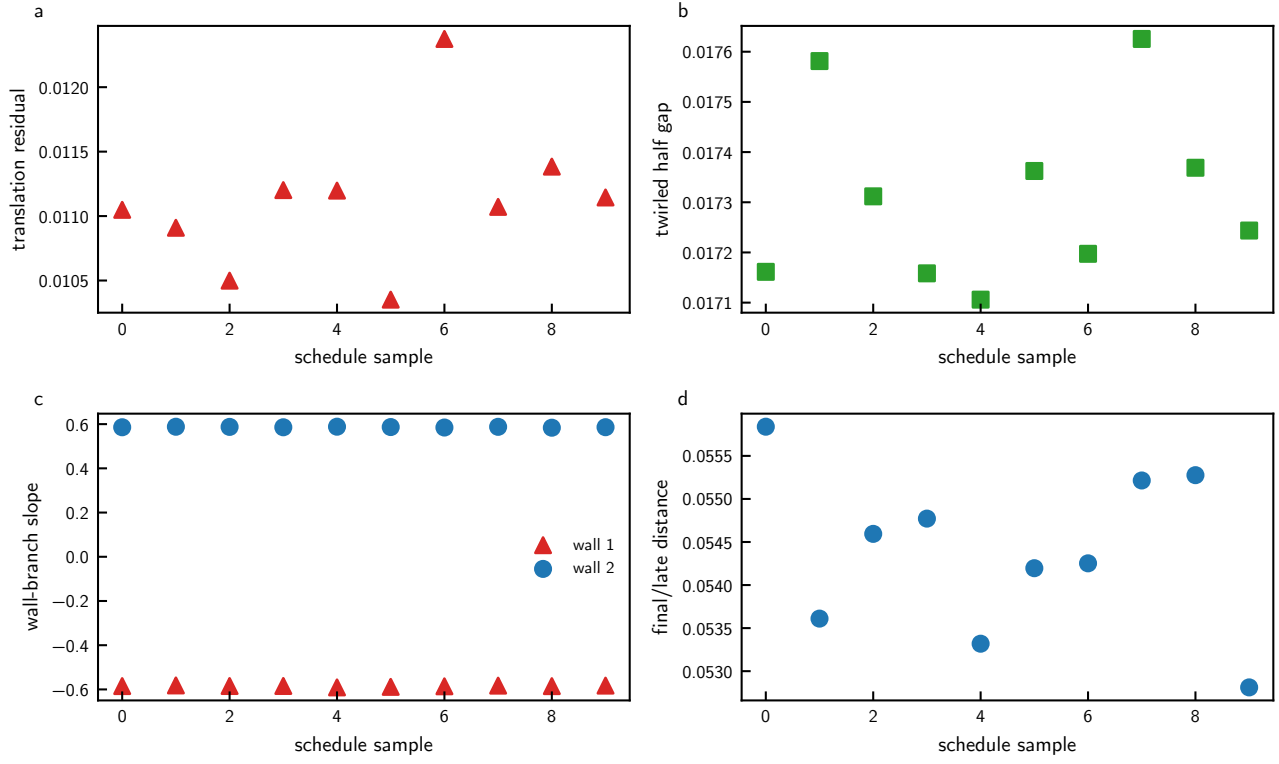


FIG. 2. Ten independent random schedules at the canonical channel endpoint. The raw translation residual ranges from 0.01035 to 0.01237, while the twirled midpoint gap is 0.01731 ± 0.00018 . Wall slopes are stable, $s_1 = -0.58513 \pm 0.00272$ and $s_2 = +0.58647 \pm 0.00152$, even though the final-to-late distance remains 0.05439 ± 0.00095 . Uncertainties are sample standard deviations over the ten schedules.